

# Day 3: Loss Functions, Cross-Entropy & MLE

Machine Learning / Deep Learning Interview Revision

**Main Theme:** Understanding loss functions, cross-entropy, softmax, and how common losses connect to Maximum Likelihood Estimation.

## 1. Loss Functions

### Question 1: What is a loss function?

**Interview Answer:** A loss function measures how far a model's prediction is from the true target. During training, we optimize the model parameters to minimize the loss.

### Question 2: What is Mean Squared Error (MSE)?

**Interview Answer:** MSE measures the average squared difference between predicted and actual values. It is commonly used for regression. Squaring the errors ensures positive and negative errors do not cancel out and gives disproportionately larger penalties to large errors.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

### Question 3: Why is MSE sensitive to outliers?

**Interview Answer:** Because the error is squared. A large error receives a disproportionately large penalty. For example, an error of 2 contributes 4 to the squared error, while an error of 10 contributes 100.

### Question 4: What is MAE and when might you use it instead of MSE?

**Interview Answer:** MAE uses the absolute difference between the prediction and the true value. Unlike MSE, its penalty grows linearly, so it is less sensitive to outliers. It can be preferred when the dataset contains significant outliers and we do not want extreme observations to disproportionately influence the model.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

## 2. Classification Losses

## Question 5: What is Binary Cross-Entropy (BCE)?

**Interview Answer:** Binary Cross-Entropy is used for binary classification and is typically combined with a sigmoid output. The model predicts the probability of the positive class, and the loss penalizes the model when it assigns a low probability to the true class.

$$\text{BCE} = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

## Question 6: What is Categorical Cross-Entropy (CCE)?

**Interview Answer:** Categorical Cross-Entropy is used for multiclass classification when each sample belongs to exactly one mutually exclusive class. Softmax produces a probability distribution over the classes, and CCE measures the negative log probability assigned to the true class.

$$\text{CCE} = -\frac{1}{n} \sum_{i=1}^n \log(p_{i, y_i})$$

## Question 7: What is Softmax?

**Interview Answer:** Softmax converts a set of logits into a probability distribution over mutually exclusive classes. Each output is between 0 and 1, and all class probabilities sum to 1.

$$p_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}$$

## Question 8: Why use Softmax for multiclass classification instead of separate Sigmoid outputs?

**Interview Answer:** In standard multiclass classification, the classes are mutually exclusive, so a sample can belong to only one class. Softmax produces probabilities that form one probability distribution and sum to 1. Separate sigmoid outputs are more appropriate for multilabel classification, where multiple classes can be true simultaneously.

## Question 9: What is the difference between binary, multiclass, and multilabel classification?

**Interview Answer:** Binary classification has two possible classes and commonly uses sigmoid with BCE. Multiclass classification has multiple mutually exclusive classes and commonly uses softmax with CCE. Multilabel classification allows multiple labels to be true at the same time and commonly uses independent sigmoid outputs with BCE.



### 3. Maximum Likelihood Estimation (MLE)

#### Question 10: What is Maximum Likelihood Estimation (MLE)?

**Interview Answer:** MLE estimates the model parameters  $\theta$  that make the observed training data as likely as possible given the inputs. In other words, we find parameter values that maximize the likelihood of observing the actual ground-truth targets.

$$\theta^* = \arg \max_{\theta} \prod_{i=1}^n P(y_i | x_i; \theta)$$

#### Question 11: What is the difference between probability and likelihood?

**Interview Answer:** Probability asks how likely the observed data is given fixed parameters. Likelihood treats the observed data as fixed and asks which parameter values make that observed data most likely. MLE uses likelihood to estimate model parameters.

#### Question 12: Why do we take the logarithm of the likelihood?

**Interview Answer:** The likelihood of a dataset is a product of many probabilities. Taking the logarithm converts the product into a sum, which is easier to compute and optimize and helps avoid numerical underflow. Because logarithm is monotonically increasing, maximizing likelihood and maximizing log-likelihood produce the same optimal parameters.

$$\log \left( \prod_i p_i \right) = \sum_i \log(p_i)$$

#### Question 13: What is Negative Log-Likelihood (NLL)?

**Interview Answer:** NLL is the negative of the log-likelihood. Since optimization is commonly formulated as minimization, maximizing likelihood is equivalent to minimizing negative log-likelihood.

$$\max L(\theta) \iff \max \log L(\theta) \iff \min [-\log L(\theta)]$$

## 4. MLE and the Derivation of Common Loss Functions

### Question 14: How does MLE lead to MSE?

**Interview Answer:** Assume that the regression target follows a Gaussian distribution around the model's predicted mean. The Gaussian likelihood contains a squared error term. Taking the negative logarithm of the likelihood produces a squared-error term. If the variance is constant, minimizing the negative log-likelihood is equivalent to minimizing MSE.

$$y_i \mid x_i \sim \mathcal{N}(\mu_{\theta}(x_i), \sigma^2)$$

Gaussian assumption  $\rightarrow$  MLE  $\rightarrow$  NLL  $\rightarrow$  MSE

### Question 15: How does MLE lead to Binary Cross-Entropy?

**Interview Answer:** For binary classification, we assume the target follows a Bernoulli distribution. The sigmoid output  $\lambda(x)$  represents  $P(y = 1 \mid x)$ . The likelihood for one sample combines both possible outcomes. Taking the negative logarithm gives the BCE loss.

$$P(y \mid x) = \lambda(x)^y [1 - \lambda(x)]^{1-y}$$

Bernoulli  $\rightarrow$  MLE  $\rightarrow$  NLL  $\rightarrow$  BCE

### Question 16: How does MLE lead to Categorical Cross-Entropy?

**Interview Answer:** For multiclass classification, we assume the target follows a categorical distribution. Softmax produces probabilities for all classes. For each sample, the likelihood contribution is the probability assigned to the true class. MLE maximizes the product of these true-class probabilities. Taking the negative logarithm gives categorical cross-entropy.

$$P(y_i = k \mid x_i; \theta) = p_{i,k}$$

Categorical  $\rightarrow$  MLE  $\rightarrow$  NLL  $\rightarrow$  CCE

### Question 17: Explain the connection between MLE, MSE, BCE, and CCE.

**Interview Answer:** MLE finds parameters  $\theta$  that maximize the likelihood of the observed ground-truth targets given the inputs. The resulting loss depends on the probability distribution assumed for the target. Under a Gaussian assumption, MLE leads to MSE. Under a Bernoulli assumption, it leads to BCE. Under a categorical assumption, it leads to CCE.

Gaussian  $\rightarrow$  MSE

Bernoulli  $\rightarrow$  BCE

Categorical  $\rightarrow$  CCE

## 5. Cross-Entropy and Classification Behavior

### Question 18: Why does cross-entropy heavily penalize confident wrong predictions?

**Interview Answer:** Cross-entropy uses the negative logarithm of the probability assigned to the true class. If the model assigns a very small probability to the correct class, the negative log becomes very large. Therefore, confidently wrong predictions receive a very high loss.

$$y = 1, \hat{y} = 0.01 \Rightarrow L = -\log(0.01) \approx 4.605$$

$$y = 1, \hat{y} = 0.99 \Rightarrow L = -\log(0.99) \approx 0.010$$

### Question 19: What happens if the model assigns exactly 0 probability to the true class?

**Interview Answer:** The logarithm of zero is undefined and approaches negative infinity. Therefore, the negative log-likelihood becomes infinitely large, causing numerical instability. In practice, probabilities are clipped to a small range or a small epsilon is added before taking the logarithm.

$$\log(0) \rightarrow -\infty$$

$$-\log(0) \rightarrow +\infty$$

### Question 20: Compare MSE, MAE, and Cross-Entropy.

**Interview Answer:** MSE is mainly used for regression and squares the prediction error, so it heavily penalizes large errors and is sensitive to outliers. MAE uses absolute error, so its penalty grows linearly and it is more robust to outliers. Cross-Entropy is used for classification and penalizes assigning low probability to the true class, especially heavily when the model is confidently wrong.

## 6. End-to-End Interview Question

**Question: Walk through what happens from a neural network producing raw outputs to calculating the final loss, and explain how this connects to MLE.**

**Interview Answer:** During the forward pass, the neural network takes an input  $x$  and produces raw logits. For multiclass classification, these logits are converted into a probability distribution over mutually exclusive classes using softmax. For each training example, we take the probability assigned to the true class as the likelihood of observing that target. MLE aims to maximize this likelihood over the training dataset. Taking the negative logarithm converts this into negative log-likelihood, which is equivalent to categorical cross-entropy. We then use gradient-based optimization such as gradient descent or Adam to minimize the loss and update the neural network parameters.

Input → Forward Pass → Logits → Softmax → Probabilities → True-Class Probability → CCE/NLL → Gradient Descent



## 7. Quick Revision Sheet

- **Regression:** Gaussian assumption  $\rightarrow$  MLE  $\rightarrow$  NLL  $\rightarrow$  MSE
- **Binary Classification:** Bernoulli assumption  $\rightarrow$  MLE  $\rightarrow$  NLL  $\rightarrow$  BCE
- **Multiclass Classification:** Categorical assumption  $\rightarrow$  MLE  $\rightarrow$  NLL  $\rightarrow$  CCE
- **Multilabel Classification:** Independent sigmoid outputs + BCE
- **Softmax:** Converts logits into probabilities that sum to 1
- **Sigmoid:** Maps a single logit to a probability between 0 and 1
- **MSE:** Large errors are heavily penalized; sensitive to outliers
- **MAE:** Linear penalty; more robust to outliers
- **Cross-Entropy:** Penalizes low probability assigned to the true class
- **MLE:** Find parameters that maximize likelihood of observed data
- **NLL:** Negative log of likelihood; minimized during training
- **Log-Likelihood:** Converts products into sums and improves numerical stability
- **Gradient Descent:** Optimizes model parameters by minimizing the loss